CARBONIC FIBER FILTER

Group: - 22232

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INTRODUCTION:

Challenges exist in everything in our life and Egypt as a developing country, tries to solve these problems. Water pollution, recycling and the existence of micro plastic of water are examples of these problems which we wish to solve them to achieve many developments in the economics of Egypt. We must analyze these problems and find the best solutions to prevent them, infecting our environment.

1- Presenting and justify the problem and solution requirements.

1. Egypt Grand Challenges
2. Problem to be solved
3. Research
4. Other solutions already tried

EGYPT GRAND CHALLENGES

- **Improve the use of alternative energies**
- **Recycle garbage and waste for economic and environmental Purposes.**
- **Deal with urban congestion and its consequences.**
- **Work to eradicate public health issues/disease.**
- **Increase the industrial and agricultural bases of Egypt.**
- **Address and reduce pollution fouling our air, water, and soil.**
- **Improve uses of arid areas.**
- **Manage and increase the sources of clean water.**
- **Deal with population growth and its consequences.**
- **Improve the scientific and technological environment for all.**
- **Reduce and adapt to the effect of climatic change**.

Recycle garbage and waste for economic and environmental purposes

Definition:

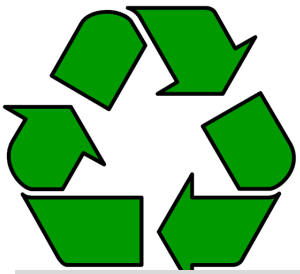


Figure (1)

It is the method of aggregation and process materials that might preferably be thrown away as trash and turning them into new product. Recycling will profit the community and also the environment. Recycling can prevent the waste of probably helpful materials and reduce the consumption of raw materials.

Between twenty plenty of "solid" waste per year, Cairo governorate occupies the very best proportion within the volume of daily waste, while use doesn't exceed 15% of the whole waste volume in Egypt consistent with the waste management organization.

Types of Recycling:

- **Waste Paper and Cardboard**



Figure (2)

Recycling paper is significant to make sure you reduce your environmental impact and reduce general waste.

- **Plastic recycling**

Most types of plastic are recyclable, and since of this, they have to be recycled to reduce the number of wastes sent to landfill and facilitate stop rubbish from ending up within the oceans.

- **Metal recycling**

To get a plan of how much energy recycling will conserve, recycling one aluminum drinks will alone, will save enough energy to power a television for around 3 hours.

- **WEEE recycling (Electronic Devices)**

WEEE recycling is for the waste of electrical and electronic equipment recycling.

• Wood utilization

Wood is that the final renewable material because of its variety of various uses.

• Glass recycling

Glass is 100 percent recyclable and never loses any purity or quality once it's recycled.

• Clothing and Textile

At least five hundredth of the textiles we tend to throw away are utile.

• Bricks and Inert Waste recycling

Hardcore rubble will be recycled into usable materials for several uses in alternative construction and building comes.

Benefits and results of Recycling:

- **Reduces the number of waste sent to landfills and incinerators.**
- **Conserves natural resources like timber, water and minerals.**
- **Increases economic security by tapping a domestic supply of materials.**
- **Prevents pollution by reducing the requirement to gather new raw materials.**
- **Saves energy.**
- **Supports Egyptian producing and conserves valuable resources.**
- **Helps produce jobs within the recycling and producing industries in Egypt.**

Steps of recycling:

Recycling includes the 3 steps below that produce a continuous loop, delineate by the acquainted recycling

• Step 1: collection and process

• Step 2: producing



Figure (3)

• **Step 3: getting new products**

made from Recycled Materials

Address and reduce pollution fouling our air, water and soil

Introduction:



Figure (4)

Pollution is that the introduction of pollutants into the natural surroundings that cause adverse changes. Pollution will take the form of chemicals or energy, like noise, heat, or light. Contaminants, constituents of contamination, is either foreign substances / energies or present pollutants. Pollution is

often classified as a degree supply or an unspecified supply of pollution. In 2015, pollution killed nine million individuals worldwide.

• Air pollution

• Plastic pollution

• Soil pollution

• Water pollution.



Figure (5)

Plastic pollution:

Plastic pollution is caused by plastic accumulation within the surroundings. Plastic, a non-biodegradable substance, is very harmful to any or all life on earth. Their lives thanks to plastic pollution.

Water pollution:

Pollution of water bodies like lakes, rivers, ponds, aquifers, etc. by pollutants is termed pollution. Like pollution, pollution is one among the most harmful types of pollution. It will have terribly serious consequences for all living organisms mistreatment contaminated water.

Air pollution:



Figure (6)

Air pollution refers to the discharge of pollutants like toxic gases, biological molecules, and stuff into the atmosphere. The pollutants is derived from many sources as well as each natural

Soil pollution:



Figure (7)

When the soil of a part is contaminated, it ends up in soil pollution or land degradation. The soil is crucial to the expansion of all plants as well as crops. Thus, a degradation within the soil quality interprets to lower yields and poor health of crops grown on such soil.

Causes & Effects of Pollution

Air Pollution

There are 2 kinds of air pollutants, primary and secondary. Primary pollutants are emitted directly from their supply, whereas secondary pollutants are fashioned once primary pollutants react within the atmosphere.

The burning of fossil fuels for transportation and electricity produces each primary and secondary pollutants and is one amongst the largest sources of pollution.

Water Pollution

Nutrient pollution is caused by waste material, sewage, and fertilizers. The high levels of nutrients in these sources find yourself in bodies of water and promote algae and weed growth, which may create the water undrinkable and depleted gas inflicting aquatic organisms to die.

Pesticides and herbicides applied to crops and residential area unites concentrate within the soil and are carried to the groundwater by rain and runoff. For these reasons anytime somebody drills a well for water it should be checked for pollutants.

Land & Soil Pollution

Land pollution is that the destruction of land as a results of human's activities and also the misuse of natural resources. this happens once humans apply chemicals like pesticides and herbicides to the soil, dispose of waste improperly, and without showing responsibility exploit minerals through mining.

Soil is additionally impure through leaky underground septic tanks, waste material systems, the action of harmful substances from landfill, and direct discharge of waste water by industrial plants into rivers and oceans.

Plastic pollution

As plastic is a smaller amount expensive, it's one amongst the most wide available and overused item within the world nowadays. once disposed of, it doesn't decompose simply and pollutes the land or air near once burned within the outside.

Effects:

Pollution has been found to be present wide within the surroundings. There are several effects of this:

• Smog and haze will reduce the number of sunlight received by plants to hold out chemical action and ends up in the assembly of tropospheric gas that damages plants.

• Soil will become unfruitful and unsuitable for plants. This may have an effect on alternative organisms within the food cycle.

• Sulfur oxide and gas oxides will cause air pollution that lowers the hydrogen ion concentration price of soil.

• Organic pollution of watercourses will consume gas levels and reduce species diversity.

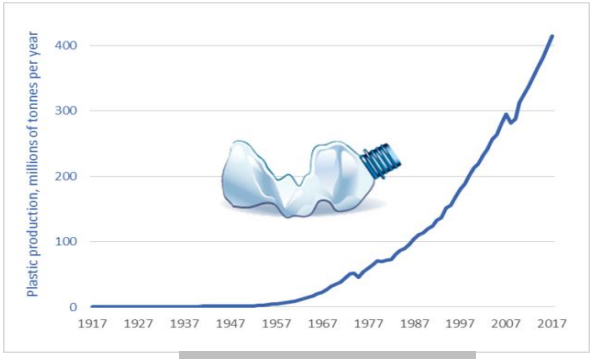
Recycle garbage and waste for economic and environmental purposes

Introduction:

Recycling is that the best way to eliminate all the materials that we tend to don't use and harm the atmosphere. Thought-about it's thought-about because the most significant method for all the countries. The wasted plastic is an example of the materials that we tend to throw it within the seas, oceans and rivers. As a results of use the wasted plastic, the pollution can decrease and lots of issues additionally are solved.

The causes of recycling:

1) The large amount of the wasted materials.
2) The pollution; that the rise of the wasted material result in increased all the categories of pollution as shown in graph (1) "the quantity of plastic increase annually.
3) Recycling reduces the requirement for raw materials.
4) Recycling help us to save lots of the energy that we tend to use a large amount of energy once creating products from raw materials.



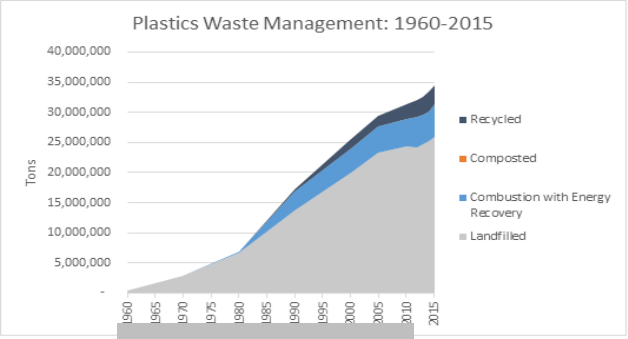
Graph (1)

The results of recycling:
1) Defend the atmosphere and every one the natural resources.
2) Save the energy and reduces the greenhouse gases emission.
3) reduce the global climate change and also the warming.
4) Recycling reduces the combustion.
5) Save fish and life from the toxicant substances.

Egypt produces annually regarding sixteen.2 million loads of waste; plastic quantity is regarding 6 June 1944 of this waste, this is often around 970 thousand loads of plastic waste annually. As a result of this vast amount of plastic wastes, we should always encourage the plastic use.

The causes of use the plastic:

1) It reduces the energy used and also the consumption of contemporary raw materials.
2) Throwing the plastics in water, harms the marine life in many ways in which and also the amount of it will increase incessantly as shown in graph (2).
3) Plastic contain some harmful substances like biphenyl A, it's harming the water badly.



Graph (2)

4) Plastic that thrown in water, take a few years to decay thus fish and life are become intoxicated.
5) It'll reduce the water and pollution.
6) It reduces the greenhouse gases emission.
7) Small plastics in water will each absorb and provides off chemicals and harmful pollutants.

"Plastics have several harmful impacts once we throw it in water. Small plastics may damage human health as they move through the marine food cycle."

The results of recycling the plastics:

1) Rescuing the marine life from plastic's harms and rescue human life as a result.
3) Decreasing the world warming and global climate change.
4) Decreasing the water and pollution.
5) Increasing the economic when recycling the wasted plastics.

What will happen if the problem solved or not?

If the problem of recycling doesn't be solved there are several unhealthy results are, such as, the number of wasted materials can increase additionally the pollution also (water and pollution specifically).
-we can use a large amount of energy to show the raw materials into new merchandise thus we are going to lose the energy stock that we tend to saved.
-climate change and warming can increase and lots of different issues additionally.
But if the matter of use be solved there are several different issues are solved additionally like pollution, saving energy, global climate change and warming.

Address and reduce pollution fouling our air, water and soil

Introduction:

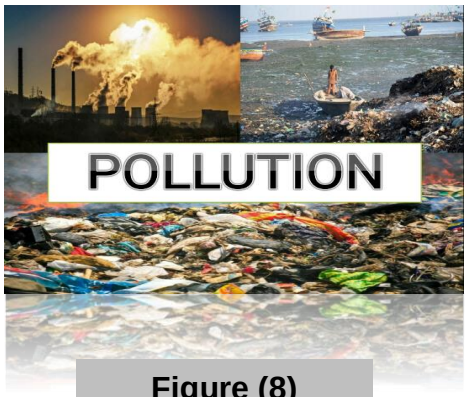


Figure (8)

Pollution is a world problem afflicting Egypt. There are several sources of pollution as shown in figure 8, like transport, deposit, accumulation, air pollution, region pollution, minerals and pollution as well as marine pollution, groundwater, wastewater, pesticides, soil pollution, sewage, sediment pollution, forest pollution, impacts of pollutants on humans. , Plants, fish, aquatic species, microorganisms, and animals. Environmental and molecular toxicity applied to pollution analysis, biosensors, climate and world changes, environmental effects of pollution models and pollution. Pollution of water, plastic, air and soil.

Plastic pollution:

Definition:



Figure (9)

Plastic pollution is any plastic that finishes up within the surroundings – from bottles and bags to less obvious sources like teabags and garments. Plastic pollution has become one among the most pressing environmental problems, as quickly increasing production

of disposable plastic products overwhelms the world's ability to take care of them.

Causes:

1. Plastic is employed nearly all over

Plastic are the most cheap and simply out there things within the current world. Plastics are low-cost and straightforward to create and are equally sturdy. They also get discarded simply.

2. Urbanization and population growth

Increasing urbanization and growth rate is significantly in command of plastic pollution. With the rise of population and concrete growth, the demand for cheaper and without delay out their materials will increase.

3. Plastics are low-cost and cheap to Manufacture

Because plastics are the cheapest and most affordable materials to manufacture, their production has tripled within the recent decades to take care of the ever-rising consumer demands.

4. Marine Shipping and Fishing industry

The shipping and fishing industry also contribute to plastic pollution, particularly within the oceans. Remote rural beaches tend to possess plastic rubbish washed to the shows that come back from the ships, ocean accidents, and from the nets used for fishing that is typically manufactured from plastic.

Results:

• Implications for Animals

Plastic wastes are mistaken for food by varied animals, principally marine life.

• Implications for Human Health

Plastics are created of a range of hepatotoxic chemicals. As such, its uses and exposure are related to variety of human health issues.

• Land Pollution

Plastics wastes have resulted within the destruction and decline in quality of the earth's land surfaces in term of use, landscape and skill to support life forms.

• Water Pollution

Many lakes and oceans have according fearful cases of plastic scrap floating on water surfaces, moving an excellent variety of aquatic creatures.

If this problem not solved:

- **Human illness**

As waste levels increase, human exposure to toxins also will increase like cancer and heart diseases.

- **Global Warming**

Burning fossil fuels for energy, like coal, oil and gas, releases CO2 and different toxins into the Earth's atmosphere.

- **Economic Effects**

Pollution, because of its ability to cause malady in humans, will have a big impact on the world's economy.

Solved:

- **Global warming can decrease**
- **Diseases caused by waste and gases are going to be reduced**
- **Agricultural crops and plants won't be destroyed**
- **Intestinal diseases like malaria and typhoid are going to be reduced**

"Microplastics problem"

- **Plastic is that the most prevalent kind of marine detritus found in our ocean and lake. Plastic detritus will are available all shapes and sizes, however those who are but 5 millimeters long (or concerning the dimensions of a sesame seed) are known as "microplastics." Microplastics are terribly little items of plastic that pollute the surroundings. Microplastics aren't a particular quite plastic, however rather any sort of plastic fragment that's but five millimeter long. They enter natural ecosystems from a range of sources, together with cosmetics, clothing, and industrial processes.**
- **Plastic is all over. Plenty of it ends up within the ocean. Most plastics within the ocean slice into terribly little particles. These little plastic bits area unit known as "microplastics." alternative plastics area unit advisedly designed to be little. They are known as small beads and are utilized in several health and wonder product. They pass unchanged through waterways into the ocean. Aquatic life and birds will mistake microplastics for food. 2 classifications of microplastics**

presently exist. Primary microplastics area unit any plastic fragments or particles that area unit already 5.0 millimeter in size or less before getting into the surroundings. These embody microfibers from article of clothing, small beads, and plastic pellets (also called nurdles). Secondary microplastics area unit microplastics that area unit created from the degradation of larger plastic product once they enter the surroundings through natural weathering processes. Such sources of secondary microplastics embody water and soda bottles, fishing nets, and plastic baggage. Each varieties area unit recognized to act the surroundings at high levels, notably in aquatic and marine ecosystems.

Primary microplastics:

• Primary microplastics are little items of plastic that are purposefully factory-made. They're typically used in facial cleansers and cosmetics, or in air blasting technology. In some cases, their use in medication as vectors for medicine was according. Microplastics "scrubbers", used in exfoliating hand cleansers and facial scrubs, have replaced historically used natural ingredients, as well as ground almonds, oatmeal, and pumice. Primary microplastics have additionally been made to be used in air blasting technology. This method involves blasting acrylic, melamine, or polyester microplastics scrubbers at machinery, engines, and boat hulls to get rid of rust and paint. As these scrubbers are used repeatedly till they diminish in size and their cutting power is lost, they usually become contaminated with serious metals like metallic element, chromium, and lead. Though several corporations have committed to reducing the assembly of small beads, there are still several bio plastic small beads that even have an extended degradation life cycle the same as traditional plastic.

Secondary microplastics:

• Secondary plastics are tiny items of plastic derived from the breakdown of larger plastic detritus, each embarrassed and onto land. Over time, the end result of physical, biological, and Che photodegradation, together with photodegradation caused by daylight exposure, will cut back the structural integrity of plastic detritus to a size that's eventually undetectable to the optic. This method of breaking down massive plastic material into a lot of smaller items is understood as fragmentation. It's thought of that microplastics would

possibly any degrade to be smaller in size, though the tiniest microplastics reportedly detected within the oceans at the present is one.6 micrometers (6.3×10−5 in) in diameter. The prevalence of microplastics with uneven shapes suggests that fragmentation may be a key supply.

• Microplastics come back from a spread of sources, together with from larger plastic dust that degrades into smaller and smaller items. Additionally, small beads, a sort of microplastics, are terribly little items of factory-made synthetic resin plastic that are extra as exfoliants to health and wonder product, like some cleansers and toothpastes. These little particles simply go through water filtration systems and find yourself within the ocean and lake, sitting a possible threat to aquatic life.

RESEARCH

- **Main topics which we have researched about:**
  - **What is microplastics?**
  - **Where is microplastics found?**
  - **The most common types of microplastics.**
  - **The ratio of microplastics in oceans, rivers and seas.**
  - **What are the different solutions which try to solve this problem (existence of microplastics in water and plastic pollution)?**

- **Topics searched related to the problem:**
  - **How to remove microplastics from water?**
  - **What are the common ways which try to remove microplastics from water?**
  - **What are the best biological and chemicals solutions to remove microplastics?**
  - **What are the common filters which remove microplastics?**

- **Topics searched related to a possible solution:**
  - **What are the main properties of activated carbon?**
  - **The main types of activated carbon**
  - **The properties of hollow fiber membrane.**
  - **The difference between granular and extruded activated carbon.**
  - **The relation between the quantity of granular carbon and the amount of removed microplastics.**

OTHER SOLUTIONS ALREADY

Ecofario

Introduction:



Figure (10)

ECOFARIO GmbH develops innovative micro plastic filtration systems supported the hydro cyclone principle to be used in municipal and industrial sewer water treatment plants. These systems square measure supposed to attenuate the small plastics entry and therefore the associated waste product input into surface waters. Thus, it ought to be secured that in the future no small plastics and no associated pollutants can reach the organic phenomenon via our sewer water.

The idea of this project:

After getting into our High-G Separators, the contaminated water is forced into significant rotation (up to 2000 G). thanks to these extraordinarily high G-forces, it's ready to separate the small plastic particles from the effluent.
ECOFARIO technology relies on the hydro cyclone principle and has been changed by North American country in such the way that it will even separate particles like small plastics, that disagree only minimally from water in terms of their density.
The small plastic particles filtered out of the wastewater, in conjunction with the adhering toxins and pollutants, square measure then place into thermal usage.

Advantages:

- **It filtered small plastic particles out of the wastewater**
- **It can also filter the adhering toxins and pollutants, are then place into thermal usage.**
- **it will be separating oil from water when oil disasters or filtering powder coat particles from industrial paint search wash waters.**

French scientists use jellyfish in the fight against micro plastics

Introduction:

Jellyfish or Plastic bag?! It's onerous to inform lately, associate degreed it has always AN unpleasant encounter either approach
But Jellyfish belong inside the ocean; plastic does not
Plus, Jellyfish might presently become our allies inside the battle against micro plastics, tiny Plastic particles polluting our waters. A study has found that secretion created by Jellyfish can capture gold nanoparticles that come from industrial processes. supported those finding, French scientists try and acquire if the secretion could in addition be used as a little plastic filter.
For this purpose, the investigator extract secretion from the Jellyfish and apply it to experimental setups inside the laboratory. If productive, the Jellyfish filters could be applied in material treatment plants, that unit a collection purpose for micro plastics. Technology for filtering micro plastics from the water throughout the treatment technique does not yet exist.
And treated water is often discharged into streams and rivers that carry the plastic to the ocean.

Advantages:

Decrease the rate of micro plastics in seawater

Disadvantages:

This way we will do these marine creatures like jellyfish

Removing Micro plastics From Water by using oil and magnetite powder

Introduction:

Magnetite could be a rock mineral and one in all the most iron ores, with the formula Fe3O4. it's one in all the oxides of iron, and is magnetic attraction, micro plastics then create their manner into waterways and area unit nearly not possible to get rid of through filtration. tiny fish area unit far-famed to eat micro plastics and as larger fish eat smaller fish these micro plastics area unit centered into larger fish species that humans consume.

The idea of this project:



Figure (11)

• employing a mixture of oil and magnetic iron-ore powder (Magnetite could be a present, non-toxic iron chemical compound.) to form a Ferro fluid within the water containing micro plastics. The micro plastics combined with the Ferro fluid that was then removed.
• once the micro plastics assured to the Ferro fluid, Ferreira used a magnet to get rid of the answer and leave solely water.

Testing the project:

After 1,000 tests, the strategy was eighty-seven effective in removing micro plastics of all kinds from water. the foremost active micro plastic removed was that from a washer with the toughest to get rid of being polyprotein plastics.

This would avoid the micro plastics from ever reaching watercourses and therefore the ocean. whereas discount within the use of micro plastics is that the ideal scenario.

Reverse Osmosis Filter

Reverse Osmosis (RO) filtration is one of the most popular water filtration methods available. In simple terms, reverse osmosis works as water is forced across a semi-permeable membrane, leaving contaminants behind that are flushed down the drain. The clean drinking water collects in a holding tank.



Figure (12)

Advantages of RO filters:

1-improve taste:

RO filtration improves taste, odor and appearance of water by removing contaminants that cause taste and odor problems.

2-simple maintenance:

RO systems have very few moving or replaceable parts make RO systems easy to clean and service.

3-*removes impurities:*

RO system removes pollutants from water including nitrates, pesticides, sulfates, fluoride, bacteria, arsenic and much more.

Disadvantages of RO systems:

1) RO water setup removes essential natural mineral like sodium, iron, calcium, and magnesium that are essential for our body and causes a scarcity of essential minerals inside the body. it's a significant disadvantage of artificial language water purifiers.

2) There is often artificial language membrane breakage. there's terribly less mechanism (water purifier) WHO inform once to modify artificial language membrane. On breakage of membrane dissolved salts, viruses and microorganism can merely undergo artificial language membrane. It's higher to modify artificial language membrane once throughout a year.

3) Ro water setup doesn't kill waterborne disease-causing microorganism and viruses. there's a high likelihood that microorganisms can endure artificial language membrane(it's higher to pass artificial language water through the ultraviolet radiation water setup to treat microorganisms)

4) Ro water setup takes too long for the purification of water. it's among the main disadvantages of artificial language water setup.

2-GENERATING AND DEFENDING A SOLUTION.

1. Design requirements

2. Selection of solution

3. Selection of prototype

SOLUTION AND DESIGN REQUIREMENTS

Solution requirement:

The solution is about a filter consisting of three stages, which is (solid carbon - carbon granules - industrial fiber). Each stage has a specific function to purify the water.

Any project has some advantages that make it distinct from other projects, which also make it successful in application in real life. Design requirements are the most important factors that determine the success of this project and also determine the applicability of this project in fact; any project must meet some design requirements, namely:-

Safety:

Safety is considered one of the most important factors for the success of projects. When conducting any project, the people who implement the prototype of the project should follow the safety instructions and not deal directly with the chemical or toxic substances. This has been followed in the implementation of the prototype of the project.

Efficiency:

In order to say that this project has a high efficiency must have the ability to purify the water from the micro-plastics; therefore, there must be a large amount of pure water from the micro-plastics out. Efficiency can be calculated by comparing between the water purity of micro-plastics before and after the filter stages.

Eco-Friendly:

When making any project, it must be taken into account that the project is eco-friendly. The materials that used in constructing the prototype must be friendly and do not cause any diseases or negative impacts on the environment, and do not use any toxic chemical substances dangerous to human health, And this has been followed in the work of the project.

Low cost:

For any project, for any project cost is one of the most important factors determining the success of the project to be applied in the real project's cost should be available for all the people. The cost of the successful project must be cheap and the materials used in it should have the same quality of the expensive materials.

Testability:

It is important that the project be tested for results to see if the project is successful or not. In addition to testing, it should be easy to know the results.

Design Requirement:

So, the design requirements which chosen are:

Safety:

Any successful project must consider the safety at first which our prototype doesn't include any pollutants to the environment.

Eco-friendly:

In order to be a successful solution the prototype shouldn't emit any harmful gases that may pollute the environment.

Efficiency:

As it is an important Comparative face between project and another as it is achieved in our prototype successfully. Our efficiency is about 80%.

PH scale:

Our prototype doesn't change the PH of the water, it is still available to drink and use after filtering it by our filter. The PH of the exit water is about 7.66.

TDS (total dissolved solids) of water:

The exit water of our filter must have the same amount or range of the TDS as the ordinary water. The TDS of exit water from our filter is about 550.

SELECTION OF SOLUTION

Our country faces a lot of problems all of these problems are grouped in 11 Egypt grand challenges, one of the problems that threaten Egypt is pollution of all kinds' especially plastic pollution and water pollution. In these years we should seek to find solutions of this problems. Micro plastic in water is one of these types of pollution which attracted the attention of researchers during last centuries. Our challenge is removing micro plastics from rivers, seas, or oceans. After searching about the risks of micro plastic it was decided to make a filter to pure the fresh water from microplastics.

Our idea summarizes in making a filter with three stages and passing the water over the all stages which the first one is extruded carbon, the second is granular carbon and the final one is hollow fiber membrane. The exit water of the filter is available to drink and use.

- We followed some steps and methods to construct our prototype they are:-

-A bottle of P.P its capacity equal 1 liter was bringing. The hollow fiber membrane was put in the bottom of the bottle with the quantity of 19.6 grams. A plate with holes of poly carbonate was located to separate between the fibers and the granular carbon. The granular Carbon was put with the quantity of 113. And placing a plate with holes of poly carbonate on it. The extruded carbon was added with the quantity of 56.3 grams and it was fixed. Thus, the filter was ready to pass the water over all its stages and water flowed to a beaker. The PH of the exit water was measured by using the PH meter and it was about 76.6, and the TDS by using the conductor device, it was about 549 ppm.

SELECTION OF PROTOTYPE

After doing a lot of research about risks of micro plastic we decided to make a filter with three stages. Each stage of it has a specific function. The components of the filter are extruded activated carbon, granular activated carbon, and hollow fiber membrane.

Prototype mechanism:

Our prototype consists of three main materials which are Extruded activated carbon, Granular Activated Carbon, and hollow fiber membrane. First, we brought a P.P bottle which doesn't pollute the environment because it doesn't react with water. After that, the materials are bought in a specific arrangement. Every stage has a specific function in which the first one is extruded activated carbon, with the quantity of 56.3 grams. It is used to absorb the impurities from water. The second one is Granular activated carbon, with the quantity of 113.4 grams. It is used to remove the undesirable components and the unwanted smell, and the final one is the fiber, about 19.6 grams, which is used to remove microplastics from water.

Extruded Activated Carbon (EAC)

- Extruded activated carbons are a cylindrical pellet product ranging in size from one mm to five mm. usually utilized in gas section reactions, EACs are a heavy-duty activated carbon as a results of the extrusion method.
- Granular activated charcoal (GAC)
- Granular activated carbons typically point particle sizes of 0.2 mm to five metric linear unit and may be utilized in each gas and liquid section applications. GACs are well-liked as a result of they provide clean handling and have a tendency to last longer than PACs. to boot, they provide improved strength (hardness) and may be regenerated and reused

Hollow Fiber Membrane:

Hollow fiber membranes are a class of artificial membranes containing a semi-permeable barrier in the form of a hollow fiber. Hollow fiber filtration utilizes thousands of long, porous filaments

ranging from 1-3.5mm wide. Each filament is very narrow in diameter and flexible. Hollow fiber can find uses in all types of filtration, ranging from microfiltration to reverse osmosis. Common applications for hollow fiber membranes include Industrial water/wastewater, juice processing, and biotech applications. Hollow fiber membranes feature a very high packing density because of the small strand diameter. Hollow fiber membranes can be a good way for filtration water from micro plastic.

Our prototype achieved the most important design requirements such as safety, high efficiency, PH scale, TDS, and eco-friendly.

Safety:

Any successful project must consider the safety at first which our prototype doesn't include any pollutants to the environment.

Eco-friendly:

In order to be a successful solution the prototype shouldn't emit any harmful gases that may pollute the environment.

Efficiency:

As it is an important Comparative face between project and another as it is achieved in our prototype successfully. Our efficiency is about 80%.

PH scale:

Our prototype doesn't change the PH of the water, it is still available to drink and use after filtering it by our filter. The PH of the exit water is about 7.66.

TDS (total dissolved solids) of water:

The exit water of our filter must have the same amount or range of the TDS as the ordinary water. The TDS of exit water from our filter is about 550.

How can we test our prototype?

By passing the water through all stages of the filter, and emptied water which the micro plastic had been removed from it in a container, we tested our prototype by using the PH meter and measuring the TDS by using the conductor device. We calculated the efficiency by using the following equation: output / input X100= 0.4/0.5= 80%

3 -CONSTRUCTING AND TESTING A PROTOTYPE.

1 Materials and Methods

2 Test plan

3 Data collection

MATERIALS

Materials Pictures					
Materials name	Bottle (P.P)	Extruded activated carbon	Granular carbon	Industrial fiber	Poly carbonate
quantity	_____	56.3 grams	113.4 grams	19.6 grams	_____
Materials cost	20 L. E	15 L. E	10 L. E	0.5 L. E	0.5 L. E
Materials source					

Table (1) shows the materials which were used

METHODS

After a continuous search about suitable materials that can be used, these steps had been followed to construct the prototype:



Figure (13)

1) We bring a bottle of P.P its capacity equal 1 liter.

2) The hollow fiber membrane was put in the bottom of the bottle with the quantity of 19.6 grams.

3) A plate with holes of poly carbonate was located to separate between the fibers and the granular carbon.

4) The granular Carbon was put with the quantity of 113 and placing a plate with holes of poly carbonate on it. 5) The extruded carbon was added with the quantity of 56.3 grams and it was fixed.

6) Thus, the filter was ready to pass the water over all its stages and water flowed to a beaker as shown in figure (11).

7) The PH of the exit water was measured by using the PH meter and it was about 76.6, and the TDS by using the conductor device, it was about 549 ppm.

TEST PLAN

Design requirement	How it can be tested
High efficiency	By calculating the output of microplastics and divided it over the input
safety	Our prototype doesn't include any pollutants to the environment.
Eco-friendly	The prototype doesn't emit any harmful gases that may pollute the environment.
PH scale	By using the PH meter the PH of the exit water=7.66
TDS	By using the conductivity device the TDS of the exit water =550 ppm

Table (2) shows the design requirements

For testing our prototype, we followed some steps which are:

1. **The filter was filled with water until it covered all its three stages.**

2. **The water passed all the three stages which consist of extruded activated carbon, granular carbon, and hollow fiber membrane.**



Figure (14)

3. **The filter was emptied of water into a container and the amount of micro plastics that the filter had removed was calculated. This was done to determine the efficiency.**

4. **To have drinkable water, the PH of water was measured by PH meter and it was about 7.6, and the TDS by using the conductor device, it was about 550 ppm as shown in figures (14) and (15).**



Figure (15)

DATA COLLECTION

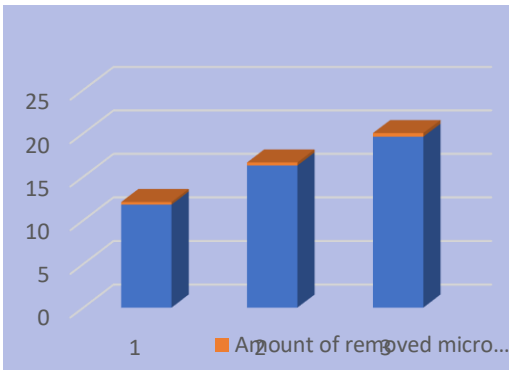
After we finished our prototype and made the test plan, we collected some negative observations and some positive results:

As for the negative observations:

The filter did not remove the micro-plastic very well from the water because there was some mistakes in the quantity of materials.

As for the positive results:

We tried to avoid all the previous mistakes as we used suitable amount of each material, after trying three different values of hollow fiber membrane as shown in table (3) and graph (3).



Graph (3)

-From the table and the graph, we concluded that there is a direct relation between the quantity of the hollow fiber membrane and the amount of removed micro plastic. "If the amount of hollow fiber membrane was increased, the amount of removed micro plastic increases."

Number of trials	Amount of hollow membrane fiber (g)	Amount of removed micro plastic (g)
First trial	11.8	0.29
Second trial	16.3	0.34
Third trial	19.6	0.42

Table (3)

-The PH of the exit water is about 7.66 and the TDS is about 550 ppm.

-As there is no measurement with 100% accurate, so we measured the error in our filter from the equation.

Relative error = |actual value – measured value|/ actual value

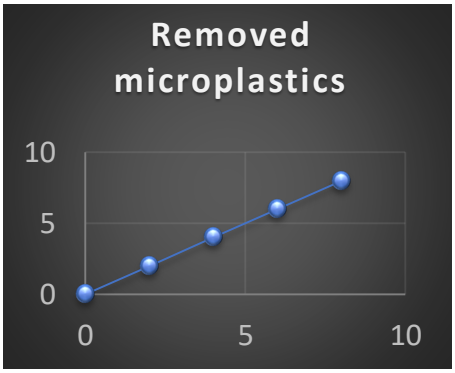
= |0.5 – 0.4| / 0.5 = 0.2

Our prototype met all the essential design requirements which are: high efficiency, eco-friendly, suitable PH and suitable TDS. By achieving these design requirements, we were able to solve this international problem that even WHO needed to overcome.

The efficiency of our filter was calculated by using the following equation:

$$\frac{\text{output (microplastics filtered)}}{\text{input(total microplastics)}} \text{ X } 100 = \frac{0.4}{0.5} \text{ x } 100 = 80\%$$

- After that we arrived to the relation between the amount of materials and the increase in the amount of removed micro plastics represent in table (4) and graph (4).If the amount of the materials increase, the amount of removed micro plastics increase, thus the efficiency remains high and constant.



Graph (4)

Number of trials	Wight of Granular Activated Carbon (g)	Wight of Extruded activated carbon (g)	Wight of hollow fiber membrane (g)	Amount of removed microplastic (g)
First trial	77.32	27.95	15.92	0.29
Second trials	95.87	40.23	17.54	0.34
Third trials	133.41	56.29	19.68	0.4

Table (4)

We get all of this result by using some measurement tools:

picture	Name	Benefit and usage
	PH meter	Measuring the PH of the water
	Conductivity device	Measuring the amount of TDS in the water.

Table (5) shows the measurement tools which were used

4- EVALUATION, REFLECTION, RECOMMENDATIONS.

1 Analysis & Discussion

2 Recommendation

3 Learning outcomes

ANALYSIS AND DISCUSSION

The analysis of the prototype behavior and the test plan results:

The prototype is a water filter, it consists of three main materials which are Extruded activated carbon, Granular Activated Carbon, and hollow fiber membrane. Every stage has a specific function, the first one contains extruded activated carbon that absorbs the impurities for its extremely porous structure that allows it to effectively capture and hold materials, and the second one contains granular activated Carbon that is used to remove undesirable components in water and any unwanted smell. The last one is hollow fiber membrane which can remove very tiny impurities like micro plastics. The prototype purifies the water from the micro plastics, by passing the water through its three stages, as it removed about 0.4 grams of the micro plastics from 0.5 grams, and we calculated the efficiency by using the following equation:
(Output/input x100) (0.4/0.5 *100 =80%).
So, our prototype meets the design requirements and very suitable to meet the criteria of the solution of the challenge of Improve sources of clean water. Also, it has the ability to hinder the problem from expansion. Furthermore, it is very suitable to be applied on a larger scale on the real life.

The result of which we collect from the test plan:

After testing our prototype, we collected some results which are:
Removing about 0.4 grams of micro plastics from 0.5 grams.
-The PH of the water is 7.66 and it was measured by using the PH meter device.
-The TDS (total dissolved solids) of the water is about 550 ppm, by using the conductivity meter device.
-The properties of water don't change and it is usable.

The conclusions that reached from the collected data:

After collecting the data, it showed that this prototype removed 0.4 grams of the micro plastics from 0.5 grams, so the efficiency is about 80% at the end. So, the prototype had passed all the tests that were put. It succeeded to remove micro plastic from water, and met the most important design requirements which are, low cost, high efficiency, and eco-friendly. As it just cost 50 pounds, our prototype efficiency is 80%, and our materials that we used don't negatively affect water. We also get result that vary between positive and negative results. From the positive result, our prototype removed 0.4 gram from 0.5 grams of total amount of micro plastic for half liter. And after testing our prototype and measuring the PH of water it was 7.67 and the TDS of water 550 PPM. From the negative result, our prototype had some mistakes in the quantities of materials and we avoided them. So, this project is very suitable to meet the criteria of the solution of the previous challenge and it has the ability to hinder the problem from expansion. Furthermore, it is very suitable to be applied on a larger scale on the real life.

RECOMMENDATION

There is no project that is completely successful. Thus, we recommend that the researchers take care about these points:

1) Try to use Hollow membrane fiber instead of industrial Fiber if it possible which it has the ability to remove the micro plastic more than any type of fibers else.

2) Use the removed Micro Plastics in the manufacture of bottles, water bottles, medicine jars and jelly jars.

LEARNING TRANSFER

Subject	L.O	Learning transfer
English	Academic writing chapter 6	Learning the academic writing, vocabulary and grammar help us to write in poster and portfolio.
Chemistry	L O 2.3	Knowing how to measure the PH and the TDS of water by PH scale and conductivity meter.
Earth science	L O 2.4	Learning more information about the layers that filter the water from the impurities.
Math	L O 2.1	Applying the law of polynomial function to make a relation on the amount of micro plastic in water.
Physics	L O 2.1	We learn the of gravitational force that from this law we can make the impurities in water to Come down to the bottom.
Computer science	L o 2.9	We learnt about programs and how to use it. We use the Photoshop to make our poster.

		And word to write our portfolio and sketch Up to make the design of our project.
Chemistry	**LO 3.1**	**Studying the difference between the acid and base and being careful about don't change the water into base or acid.**
Computer science	**L O 2.2**	**Studying "Do while loop" in the programming language C++ and having the alternative ideas.**
Earth science	**LO 3.1**	**We use concept of conservation bases on the consumptive and no consumptive uses so we chose subsurface irrigation to increase the non-consumptive uses because it reduces the evaporation.** **We know the water treatment levels such as Screening, Flocculation, Settling and filtration.**
Biology	**LO 1.1**	**We learnt about The GMO (Genetically modified organisms) that helped us understand the prior solutions of modifying the bacteria and make it eat microplastics.**

Table (6) shows the learning transfer

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Thanks for Reading